통계열역학 (2013년도 1학기)

과제 #1 정답 (**총 110 점**)

1. **(총 20점, 각 10점)**

(a)

$$\begin{aligned} q_{elec}(T) &= \sum_{i=1} g_{ei} e^{-\beta\epsilon_{ei}} = \sum_{i=1} g_{ei} \exp\left[-\frac{hc\bar{\nu}}{k_B T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-\frac{(6.62608\times10^{-34}\ \text{J s})(2.99792458\times10^{10}\ \text{cm s}^{-1})}{(1.38066\times10^{-23}\ \text{J K}^{-1})} \cdot \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right] \\ &= \sum_{i=1} g_{ei} \exp\left[-(1.439\ \text{cm K}^{-1}) \frac{\bar{\nu}(\text{cm}^{-1})}{T} \right]. \end{aligned}$$

위 식에서 g_{ei} 는 각 에너지 준위에서 축퇴도이고 ϵ_{ei} 는 i 번째 에너지 준위의 에너지이다. 편의상 $\epsilon_{e1} = 0$ 이라고 한다.

$$\begin{aligned} q(300\ \text{K}) &= 5 + 1e^{-1.439*5000/300} + 3e^{-1.439*10000/300} = 5.00 \\ q(5000\ \text{K}) &= 5 + 1e^{-1.439*5000/5000} + 3e^{-1.439*10000/5000} = 5.41 \end{aligned}$$

(b) 원자가 바닥상태 (에너지 준위 1)와 첫 번째 들뜬상태 (에너지 준위 2)에 존재할 확률은 각각

$$p_{ei} = \frac{g_{ei} e^{-\beta\epsilon_{ei}}}{q_{\text{elec}}(T)}.$$

온도	p_1	p_2	p_3
(i) 300 K	1.00	0.00	0.00
(ii) 5000 K	0.925	0.0439	0.0312

2. **(총 20점)**

(a)

$$\begin{aligned} p_{\text{bound}} &= \frac{(c/c_0)e^{\beta\epsilon_{\text{sol}}}}{e^{\beta\epsilon_{\text{b}}} + (c/c_0)e^{\beta\epsilon_{\text{sol}}}} \\ &= \frac{(c/c_0)e^{-\beta(\epsilon_{\text{b}}-\epsilon_{\text{sol}})}}{1 + (c/c_0)e^{-\beta(\epsilon_{\text{b}}-\epsilon_{\text{sol}})}} = \frac{(c/c_0)e^{-\beta\Delta\epsilon}}{1 + (c/c_0)e^{-\beta\Delta\epsilon}} \end{aligned} \tag{1}$$

위 식에서 $\Delta\epsilon = \epsilon_{\text{b}} - \epsilon_{\text{sol}}$.

(b) ligand 농도에 따른 p_{bound}

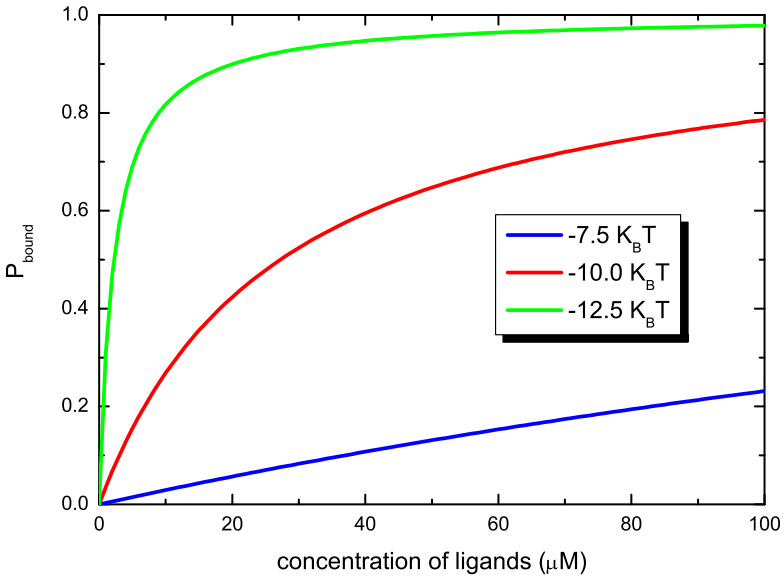


그림 1 – ligand 농도에 따른 p_{bound}.

3. **(총 70점, 각 10점)**

(a) Ligand-receptor 정준분배함수 $Q(L,V,T)$:

$$
\begin{aligned}
Q(L,V,T) = & \underbrace{\frac{\Omega!}{L!(\Omega-L)!}e^{-\beta L\epsilon_{sol}}}_{\text{상태 1}} + \underbrace{\frac{\Omega!}{(L-1)!(\Omega-(L-1))!}e^{-\beta(\epsilon_b+(L-1)\epsilon_{sol})}}_{\text{상태 2}} \\
& + \underbrace{\frac{\Omega!}{(L-1)!(\Omega-(L-1))!}e^{-\beta(\epsilon_b+(L-1)\epsilon_{sol})}}_{\text{상태 3}} + \underbrace{\frac{\Omega!}{(L-2)!(\Omega-(L-2))!}e^{-\beta(2\epsilon_b+J+(L-2)\epsilon_{sol})}}_{4} \\
\approx & \frac{\Omega^L}{L!}e^{-\beta L\epsilon_{sol}} + \frac{\Omega^{L-1}}{(L-1)!}e^{-\beta(\epsilon_b+(L-1)\epsilon_{sol})} \\
& + \frac{\Omega^{L-1}}{(L-1)!}e^{-\beta(\epsilon_b+(L-1)\epsilon_{sol})} + \frac{\Omega^{L-2}}{(L-2)!}e^{-\beta(2\epsilon_b+J+(L-2)\epsilon_{sol})} \\
= & \frac{\Omega^L}{L!}e^{-\beta L\epsilon_{sol}} + \frac{2\Omega^{L-1}}{(L-1)!}e^{-\beta(\epsilon_b+(L-1)\epsilon_{sol})} + \frac{\Omega^{L-2}}{(L-2)!}e^{-\beta(2\epsilon_b+J+(L-2)\epsilon_{sol})} \qquad (2)
\end{aligned}
$$

(b) $p_0 \leftrightharpoons$

$$
\begin{aligned}
p_0 &= \frac{\Omega^L}{L!} e^{-\beta L \epsilon_{sol}} / Q(L,V,T) \\
&= \frac{\frac{\Omega^L}{L!} e^{-\beta L \epsilon_{sol}}}{\frac{\Omega^L}{L!} e^{-\beta L \epsilon_{sol}} + \frac{2\Omega^{L-1}}{(L-1)!} e^{-\beta(\epsilon_b + (L-1)\epsilon_{sol})} + \frac{\Omega^{L-2}}{(L-2)!} e^{-\beta(2\epsilon_b + J + (L-2)\epsilon_{sol})}} \\
&= \frac{1}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L(L-1)}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&\approx \frac{1}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L^2}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{1}{1 + (2c/c_0) e^{-\beta(\epsilon_b - \epsilon_{sol})} + (c/c_0)^2 e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{1}{1 + (2c/c_0) e^{-\beta \Delta\epsilon} + (c/c_0)^2 e^{-\beta(2\Delta\epsilon + J)}}
\end{aligned} \tag{3}
$$

(c) $p_1 \leftrightharpoons$

$$
\begin{aligned}
p_1 &= \frac{2\Omega^{L-1}}{(L-1)!} e^{-\beta(\epsilon_b + (L-1)\epsilon_{sol}} / Q(L,V,T) \\
&= \frac{2\frac{\Omega^{L-1}}{(L-1)!} e^{-\beta(\epsilon_b + (L-1)\epsilon_{sol}}}{\frac{\Omega^L}{L!} e^{-\beta L \epsilon_{sol}} + \frac{2\Omega^{L-1}}{(L-1)!} e^{-\beta(\epsilon_b + (L-1)\epsilon_{sol})} + \frac{\Omega^{L-2}}{(L-2)!} e^{-\beta(2\epsilon_b + J + (L-2)\epsilon_{sol})}} \\
&= \frac{\frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})}}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L(L-1)}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&\approx \frac{\frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})}}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L^2}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{(2c/c_0) e^{-\beta(\epsilon_b - \epsilon_{sol})}}{1 + (2c/c_0) e^{-\beta(\epsilon_b - \epsilon_{sol})} + (c/c_0)^2 e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{(2c/c_0) e^{-\beta \Delta\epsilon}}{1 + (2c/c_0) e^{-\beta \Delta\epsilon} + (c/c_0)^2 e^{-\beta(2\Delta\epsilon_b + J)}}
\end{aligned} \tag{4}
$$

(d) p_2 는

$$
\begin{aligned}
p_2 &= \frac{\Omega^{L-2}}{(L-2)!} e^{-\beta(2\epsilon_b + J + (L-2)\epsilon_{sol}} / Q(L,V,T) \\
&= \frac{\frac{\Omega^{L-2}}{(L-2)!} e^{-\beta(2\epsilon_b + J + (L-2)\epsilon_{sol}}}{\frac{\Omega^L}{L!} e^{-\beta L \epsilon_{sol}} + \frac{2\Omega^{L-1}}{(L-1)!} e^{-\beta(\epsilon_b + (L-1)\epsilon_{sol})} + \frac{\Omega^{L-2}}{(L-2)!} e^{-\beta(2\epsilon_b + J + (L-2)\epsilon_{sol})}} \\
&= \frac{\frac{L(L-1)}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L(L-1)}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&\approx \frac{\frac{L^2}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}}{1 + \frac{2L}{\Omega} e^{-\beta(\epsilon_b - \epsilon_{sol})} + \frac{L^2}{\Omega^2} e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{(c/c_0)^2 e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}}{1 + (2c/c_0) e^{-\beta(\epsilon_b - \epsilon_{sol})} + (c/c_0)^2 e^{-\beta(2\epsilon_b + J - 2\epsilon_{sol})}} \\
&= \frac{(c/c_0)^2 e^{-\beta(2\Delta\epsilon + J)}}{1 + (2c/c_0) e^{-\beta\Delta\epsilon} + (c/c_0)^2 e^{-\beta(2\Delta\epsilon + J)}}
\end{aligned} \tag{5}
$$

(e) $J = 0$ 인 경우에, (5) 식은

$$
\begin{aligned}
p_2 &= \frac{(c/c_0)^2 e^{-2\beta\Delta\epsilon}}{1 + (2c/c_0) e^{-\beta\Delta\epsilon} + (c/c_0)^2 e^{-2\beta\Delta\epsilon}} \\
&= \frac{(c/c_0)^2 e^{-2\beta\Delta\epsilon}}{\{1 + (c/c_0) e^{-\beta\Delta\epsilon}\}^2} = \frac{\{(c/c_0) e^{-\beta\Delta\epsilon}\}^2}{\{1 + (c/c_0) e^{-\beta\Delta\epsilon}\}^2} \\
&= \left(\frac{(c/c_0) e^{-\beta\Delta\epsilon}}{1 + (c/c_0) e^{-\beta\Delta\epsilon}} \right)^2 = p_{\text{bound}}^2
\end{aligned} \tag{6}
$$

(f) $J = 0$ 일 때,

(g) $J = -2.5 k_B T$ 일 때,

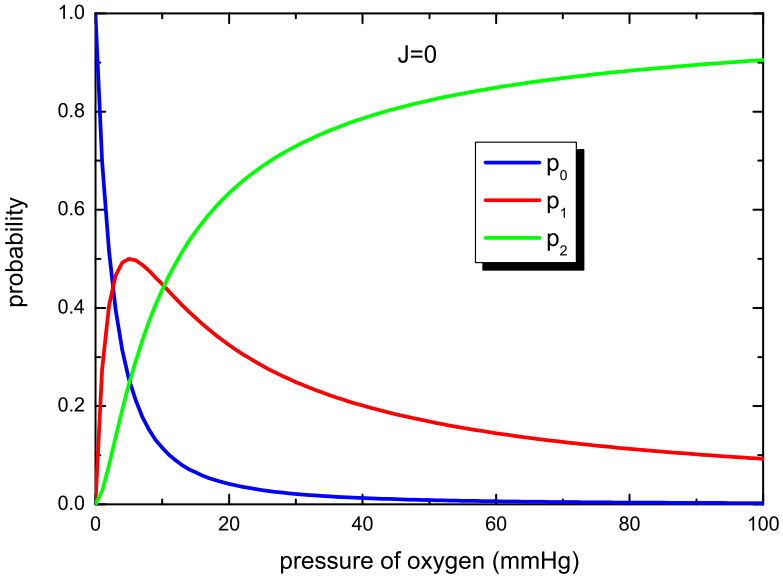


그림 2 – $J = 0$인 경우에 O_2 압력에 따른 p_1, p_2, 그리고 p_3.

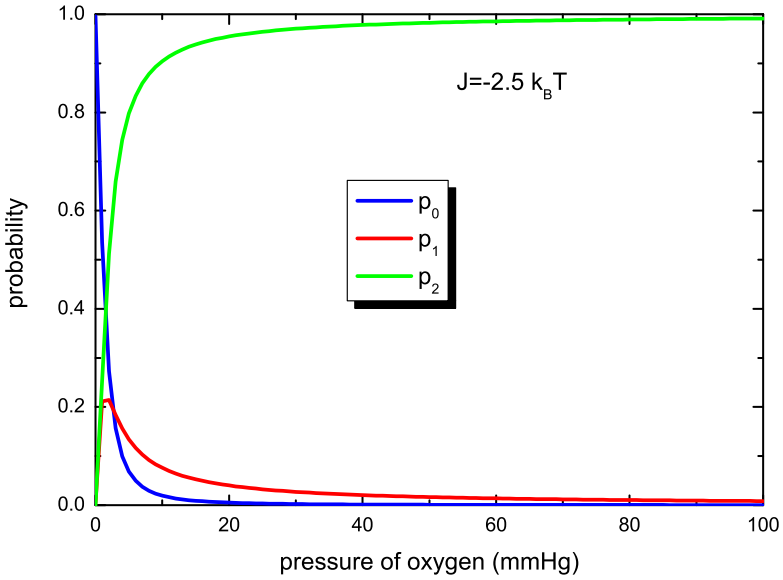


그림 3 – $J = -2.5k_BT$인 경우에 O_2 압력에 따른 p_1, p_2, 그리고 p_3.